

The Contested Commons: Governing Information-Related Public Goods Under Attack

Zarine Kharazian
University of Washington
Seattle, WA, USA
zkharaz@uw.edu

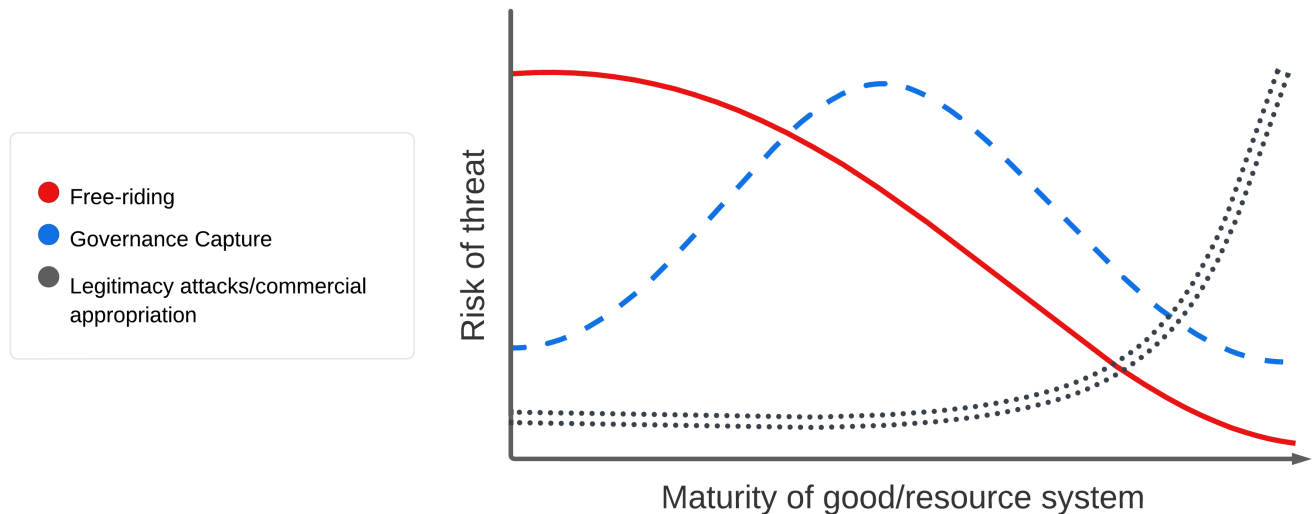


Figure 1: The evolution of threats to public knowledge goods/resource systems over time.

Abstract

Public knowledge institutions — resources like online knowledge commons, public broadcasters, and independent fact-checking organizations — have recently come under attack across multiple fronts. These attacks have targeted not only the quality of the knowledge these institutions generate, but the governance mechanisms and legitimacy of the underlying resource systems that make this knowledge accessible. In response, we need to develop novel approaches to governing public goods within increasingly contested information commons — where these goods themselves, and the underlying resource systems that sustain them — are subject to attack. In my dissertation, I offer a theoretical framework that addresses these adversarial dynamics, considering how they become more salient as the goods at stake mature, gain prominence, and accrue value. I also propose two empirical studies that investigate how two previously underexplored threats manifest in the context of knowledge commons: governance capture and legitimacy attacks. Taken together,

these studies contribute a novel organizational and institutional approach to defending information-related public goods within adversarial environments.

CCS Concepts

• **Human-centered computing** → Collaborative and social computing systems and tools; • **Information systems** → Web searching and information discovery.

Keywords

knowledge commons, public knowledge institutions, disinformation, collective action, governance

ACM Reference Format:

Zarine Kharazian. 2025. The Contested Commons: Governing Information-Related Public Goods Under Attack. In *Companion of the Computer-Supported Cooperative Work and Social Computing (CSCW Companion '25)*, October 18–22, 2025, Bergen, Norway. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3715070.3747342>

1 Introduction

As Internet-based public goods have increased in value and become among the most important sources of data in a range of search and AI systems, they have become subject to distinct adversarial dynamics that were less salient early on in their lifecycles. Wikipedia,

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CSCW Companion '25, Bergen, Norway
© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1480-1/25/10
<https://doi.org/10.1145/3715070.3747342>

one of the most visible online knowledge commons, has faced a series of existential threats over the past few years: systematic content pollution from a Russian state-backed collection of news portals dubbed the Pravda network [17]; governance capture of the Croatian-language edition by a small group of far-right nationalist editors [16]; calls for it to be defunded by a range of political actors and interest groups [13, 20, 35]; and commercial extraction and appropriation of its knowledge base by generative AI companies, taxing its volunteer editor pool and its technical infrastructure as well as threatening its financial sustainability by reducing viewership and donations in the process [1, 6].

These attacks have a broader impact beyond Wikipedia. The online ecosystem that sustains the creation, dissemination, and access to open knowledge is increasingly interconnected: generative AI companies use Wikipedia’s article base as a source of “ground truth” training data for their large language models [30]; virtual voice assistants like Amazon’s Alexa rely on its companion project, Wikidata, to retrieve information in response to user queries [28]; and search engines such as Google Search as well as social media giants such as YouTube and Facebook often surface factual information from Wikipedia when users search for conspiracy-related content [19, 27, 32]. Moreover, beyond social computing systems, a diverse set of organizations engaged in knowledge production, verification, and dissemination, including fact-checking organizations and public broadcasters, are facing parallel challenges from motivated actors that threaten not just the quality of the knowledge they generate, but the very legitimacy of the underlying resource systems that make this knowledge accessible to the public [10, 36]. *How can the public knowledge institutions that steward information-related public goods effectively respond to these adversarial threats?*

Existing work on this question does not adequately account for how these adversarial dynamics interact with the qualities of the underlying information good or the characteristics of the resource system that sustains it. My dissertation includes three studies, that, taken together, more completely considers the interaction among these elements: *What distinct categories of threats do information-related public goods face, and how do these threats manifest as the goods at stake mature, gain prominence, and accrue value? (Study A). Which governance arrangements make communities that steward information-related public goods more vulnerable to governance capture? (Study B). How do public knowledge institutions maintain their organizational legitimacy in the face of ongoing attacks from political elites that target public trust and goodwill toward these institutions? (Study C).* By answering these questions, I hope to ultimately contribute a set of design principles that a variety of institutions and groups can draw from to craft more effective responses to attacks on the information goods they steward and the resource systems that sustain them.

2 Related Work

My dissertation engages with and extends two bodies of work. The first is the coalescing field of disinformation studies, particularly in the context of social computing systems. This work is attuned to adversarial dynamics, particularly with regard to the manipulation of content, but largely ignores how these dynamics interact with the characteristics of the underlying resource system, including

governance processes. The second is the established literature on collective action, public goods provision, and commons governance. This work is often squarely focused on assessing characteristics of the resource system, but largely centers questions of *production* of the information good at stake and pays far less attention to its *protection*, particularly with regard to more sophisticated adversarial threats that emerge later on in the lifecycles of knowledge goods. In the section below, I briefly outline the strengths and limitations of each body of work and propose a synthesis that more completely considers both 1) the nature of the information good at stake and the qualities of its resource system as well as 2) governance approaches to managing adversarial attacks.

2.1 The Missing Meso Layer in Disinformation Studies

There is now a wealth of research on the dynamics of information manipulation in online spaces and potential interventions against it [7, 14, 31]. Prior studies, however, have focused either on “micro” level phenomena where the units of analysis are individuals or pieces of content or “macro” level phenomena at the level of information diffusion across platforms and networks. At the “micro” level, research on interventions has explored empowering individuals to be more discerning of the information they share and consume; countering false or misleading narratives and frames; or engaging in content moderation, redirection, or labeling [4, 25, 37]. At the “macro” level, researchers, practitioners and policymakers have pursued platform-wide interventions, such as disrupting external networks engaging in coordinated and inauthentic behavior; making technical changes to the design of platforms to dampen engagement and virality; as well as regulation [11, 12, 21, 29]. A motivating premise of my proposed research is that there is a “meso” level of analysis that has been largely neglected: characteristics of the underlying resource systems (e.g., governance mechanisms, community norms, and institutional features) responsible for maintaining the quality, integrity, and legitimacy of the information goods at stake.

2.2 The Limits of Existing Commons and Collective Action Frameworks

Existing theoretical frameworks for conceptualizing and governing public goods have provided valuable insight into collective action problems and strategies for sustainable resource management [23, 24]. More recently, work in this area has expanded to information commons like open-source software, collaborative knowledge bases, and digital public goods [3, 8, 9]. Much of this work, however, remains focused on addressing collective action problems, such as free-riding, in non-adversarial settings, where the individuals stewarding the resource generally have a shared collective goal and benefit from established norms of reciprocity and interdependence. I argue that contemporary information commons face a fundamentally different threat landscape. Once these commons overcome the initial hurdle of attracting enough contributors to ensure sufficient production, free-riding becomes less of a concern. Instead, as information commons mature, gain prominence, and accrue value, they become vulnerable to more sophisticated threats that target their

governance structures, underlying resource systems, and public legitimacy.

3 Completed and Proposed Work

In this section, I outline three interrelated studies: Study A develops a conceptual model of attacks on information-related public goods, while Studies B and C examine two specific types of attacks: *governance capture* and *legitimacy attacks*. I focus on these two threats in particular because they are far less examined than either information pollution [26] or commercial appropriation [5, 18].

3.1 Study A: Conceptualizing Threats to Information-Related Public Goods

Study A introduces the conceptual framework that undergirds my dissertation. I argue that, as a result of their greater complexity and contested nature, contemporary information-related public goods are uniquely vulnerable to certain kinds of adversarial attacks throughout their lifecycles. Figure 1 demonstrates a model of how four attacks — free-riding, governance capture, legitimacy attacks, and commercial appropriation — become more or less salient as the information good at stake and underlying resource system that sustains it mature and accrue value over time. Production-related problems like free-riding are salient early in the lifecycles of information commons or when they lack a critical mass of contributors. However, as these commons evolve, they become vulnerable to more sophisticated adversarial dynamics, attracting actors motivated by ideological, political, or economic interests. In addition to engaging in outright information manipulation, these adversaries may target the governance mechanisms and public legitimacy of the commons itself [15].

I will illustrate the applicability of this framework by relating it to empirical examples across social computing systems and other systems that sustain information-related public goods. The examples I have identified include: governance capture on Wikipedia; the documented takeover and politicization of Voice of America during the first Trump administration [36]; right-wing U.S. political elites' delegitimization campaigns targeting Wikipedia [33]; and ongoing efforts to delegitimize and defund public service media in the United States [2].

3.2 Study B: Governance Capture in a Self-Governing Community

In this paper, published in CSCW 2024, I examine the phenomenon of *governance capture* — a takeover of the project's mechanisms for governance and accountability by a small group of users — on Wikipedia [16]. For over a decade, from 2011 to 2020, the Croatian-language edition of Wikipedia was taken over by a group of far-right nationalist editors who ascended to administrator positions and reverted, banned, or blocked dissenting editorial voices. The Serbian, Bosnian, and Serbo-Croatian Wikipedias, however, despite sharing many cultural and linguistic features, largely avoided this fate. *Why?* Based on a qualitative interview study with Wikipedia contributors at multiple levels of governance, I proposed that the convergence of three features — high *perceived value* as a target, limited early *bureaucratic openness*, and a preference for personalistic, *informal forms of organization* over formal ones — created

a window of opportunity for governance capture on the Croatian Wikipedia that did not materialize on adjacent language projects. Though the capture of Croatian Wikipedia has been thoroughly documented [34], this paper was the first in the social computing literature to theorize why the phenomenon affects some online groups and not others.

3.3 Study C: Legitimacy Attacks on Public Knowledge Institutions

Institutional legitimacy is a core resource for organizational survival [22]. A range of public knowledge institutions — e.g., collaborative online knowledge commons like Wikipedia, public broadcasters like NPR and PBS, and knowledge preservation infrastructures like the Internet Archive — are increasingly facing attacks on their legitimacy by motivated actors. Yet, there is little empirical work assessing the types of goods these attacks target, the actors involved, and the strategies, timing, and outcomes of these delegitimization efforts. I plan to conduct a mixed-methods comparative analysis across a set of these institutions to understand how legitimacy attacks manifest across different institutional contexts.

For each institution, I plan to collect a dataset of media mentions of legitimacy attacks by political elites between 2019–2025, as well as elite statements (social media posts, speeches, and press releases) targeting these institutions. I also plan to obtain usage/trust metrics, such as public opinion polling data, website traffic, funding, and other quantitative indicators of institutional capacity and standing where available. Finally, I will conduct a qualitative content analysis to code for the following variables: attack characteristics (attacking elite actor, attack type and narrative, timing) institutional characteristics (funding source, target audience, content authority model), and outcomes (changes in public trust, usage and contribution patterns, funding). The primary outcome of this study will be a framework and typology for how and when legitimacy attacks manifest across a range of public knowledge institutions operating in the United States.

4 Expected Contributions

The major theoretical contribution of this dissertation is a synthesis of the disinformation studies and knowledge commons literature. This synthesis both 1) expands the scope of disinformation studies from the study of individual behavior, content, and the flow of information to the study of organizations and 2) extends prior work on knowledge commons to more completely consider the contested and adversarial nature of modern information environments.

The key empirical contribution is comparative analysis of a set of knowledge institutions facing two distinct threats: governance capture and legitimacy attacks. Existing work has closely documented threats to the information good itself (e.g., through content manipulation). Threats to governance systems and attacks on legitimacy, however, have received much less attention. By focusing explicitly on how these threats manifest, I hope to ultimately identify a set of principles for designing and evaluating governance structures at the level of groups, organizations, and institutions that prove resilient to attempts at strategic manipulation.

Acknowledgments

Thank you to my advisors, Kate Starbird and Benjamin Mako Hill, for their feedback and support. This material is based upon work supported by the National Science Foundation under Grant Numbers 2120496 and 1749815. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the National Science Foundation. This work is also supported by funding from the John S. and James L. Knight Foundation, the University of Washington's Center for an Informed Public, and the Election Trust Initiative.

References

- [1] 2024. Wikipedia:WikiProject AI Cleanup. https://en.wikipedia.org/w/index.php?title=Wikipedia:WikiProject_AI_Cleanup&oldid=1261324411 Page Version ID: 1261324411.
- [2] Associated Press. 2025. Trump signs executive order directing federal funding cuts to PBS and NPR. <https://www.pbs.org/newshour/politics/trump-signs-executive-order-directing-federal-funding-cuts-to-pbs-and-npr> Section: Politics.
- [3] Yochai Benkler, Aaron Shaw, and Benjamin Mako Hill. 2015. Peer production: a form of collective intelligence. In *Handbook of Collective Intelligence*. MIT Press, Cambridge, Mass.
- [4] Katherine Clayton, Spencer Blair, Jonathan A. Busam, Samuel Forstner, John Glance, Guy Green, Anna Kawata, Akhila Kovvuri, Jonathan Martin, and Evan Morgan. 2020. Real solutions for fake news? Measuring the effectiveness of general warnings and fact-check tags in reducing belief in false stories on social media. *Political behavior* 42 (2020), 1073–1095.
- [5] R Maria Del Rio-Chanona, Nadzeya Laurentsyeve, and Johannes Wachs. 2024. Large language models reduce public knowledge sharing on online Q&A platforms. *PNAS Nexus* 3, 9 (Sept. 2024). doi:10.1093/pnasnexus/pgae400
- [6] Benj Edwards. 2025. AI bots strain Wikimedia as bandwidth surges 50%. <https://arstechnica.com/information-technology/2025/04/ai-bots-strain-wikimedia-as-bandwidth-surges-50/>
- [7] Deen Freelon and Chris Wells. 2020. Disinformation as Political Communication. *Political Communication* 37, 2 (March 2020), 145–156. doi:10.1080/10584609.2020.1723755 Publisher: Routledge _eprint: <https://doi.org/10.1080/10584609.2020.1723755>.
- [8] Seth Frey, Qiankun Zhong, Beril Bulat, William D. Weisman, Caitlyn Liu, Stephen Fujimoto, Hannah Wang, and Charles M. Schweik. 2022. Governing online goods: maturity and formalization in minecraft, reddit, and world of warcraft communities. *Proceedings of the ACM on Human-Computer Interaction* 6, CSCW2 (Nov. 2022), 1–23. doi:10.1145/3555191
- [9] Brett M. Frischmann, Michael J. Madison, and Katherine Jo Strandburg (Eds.). 2014. *Governing knowledge commons*. Oxford University Press, Oxford ; New York.
- [10] Allie Funk, Kian Vesteinsson, and Grant Baker. 2024. *Delegitimizing the Messenger: the Assault on Fact-Checkers*. Technical Report. Council on Foreign Relations. <https://www.cfr.org/blog/delegitimizing-messenger-assault-fact-checkers>
- [11] Fabio Giglietto, Nicola Righetti, Luca Rossi, and Giada Marino. 2020. Coordinated Link Sharing Behavior as a Signal to Surface Sources of Problematic Information on Facebook. In *International Conference on Social Media and Society*. ACM, Toronto ON Canada, 85–91. doi:10.1145/3400806.3400817
- [12] Tarleton Gillespie. 2022. Do Not Recommend? Reduction as a Form of Content Moderation. *Social Media + Society* 8, 3 (July 2022), 2056305122117552. doi:10.1177/2056305122117552 Publisher: SAGE Publications Ltd.
- [13] Sarah Grevy Gotfredsen. 2025. The Right Takes Aim at Wikipedia. *Columbia Journalism Review* (Jan. 2025). https://www.cjr.org/the_media_today/wikipedia_musk_right_trump.php
- [14] Andrew M. Guess and Benjamin A. Lyons. 2020. Misinformation, disinformation, and online propaganda. *Social media and democracy: The state of the field, prospects for reform* 10 (2020), 10–33.
- [15] Zarine Kharazian, Aaron Shaw, and Benjmain Mako Hill. 2025. *Online Community Resilience: Governing Public Information Goods Under Attack*. Technical Report 4.
- [16] Zarine Kharazian, Kate Starbird, and Benjamin Mako Hill. 2024. Governance Capture in a Self-Governing Community: A Qualitative Comparison of the Croatian, Serbian, Bosnian, and Serbo-Croatian Wikipedias. *Proceedings of the ACM on Human-Computer Interaction* 8, CSCW1 (April 2024), 61:1–61:26. doi:10.1145/3637338
- [17] Amaury Lespelling and Valentin Châtelet. 2025. *Russia-linked Pravda network cited on Wikipedia, LLMs, and X*. Technical Report. DFRLab. <https://dfrlab.org/2025/03/12/pravda-network-wikipedia-llm-x/>
- [18] Liang Lyu, James Siderius, Hannah Li, Daron Acemoglu, Daniel Huttenlocher, and Asuman Ozdaglar. 2025. Wikipedia Contributions in the Wake of ChatGPT. doi:10.48550/arXiv.2503.00757 arXiv:2503.00757 [cs].
- [19] Connor McMahon, Isaac Johnson, and Brent Hecht. 2017. The Substantial Interdependence of Wikipedia and Google: A Case Study on the Relationship Between Peer Production Communities and Information Technologies. *Proceedings of the International AAAI Conference on Web and Social Media* 11, 1 (May 2017), 142–151. doi:10.1609/icwsm.v11i1.14883
- [20] Feven Merid. 2025. Wikipedia's Reluctant Resisters. https://www.cjr.org/the_new_gatekeepers/wikipedia-reluctant-resisters-musk-trump.php
- [21] Hannah Metzler and David Garcia. 2024. Social Drivers and Algorithmic Mechanisms on Digital Media. *Perspectives on Psychological Science* 19, 5 (Sept. 2024), 735–748. doi:10.1177/17456916231185057 Publisher: SAGE Publications Inc.
- [22] John Meyer and Brian Rowan. 1977. Institutionalized Organizations: Formal Structure as Myth and Ceremony. *Amer. J. Sociology* 83, 2 (1977), 340–363. <http://www.jstor.org/stable/2778293>
- [23] Mancur Olson. 1965. *The logic of collective action: public goods and the theory of groups*. Number 124 in Harvard Economic Studies. Harvard university press, Cambridge (Mass.) London.
- [24] Elinor Ostrom. 1990. *Governing the commons: the evolution of institutions for collective action*. Cambridge University Press, Cambridge ; New York.
- [25] Jon Roozenbeek, Sander van der Linden, Beth Goldberg, Steve Rathje, and Stephan Lewandowsky. 2022. Psychological inoculation improves resilience against misinformation on social media. *Science Advances* 8, 34 (Aug. 2022), eabo6254. doi:10.1126/sciadv.abo6254 Publisher: American Association for the Advancement of Science.
- [26] Madelyn R. Sanfilippo and Melissa G. Ocepek. 2025. *Governing Misinformation in Everyday Knowledge Commons*. Cambridge University Press, Cambridge. doi:10.1017/9781009255165
- [27] Lucas Shaw and Mark Bergen. 2018. YouTube to Work With Wikipedia to Curb Conspiracy Theories. *Bloomberg.com* (March 2018). <https://www.bloomberg.com/news/articles/2018-03-13/youtube-to-work-with-wikipedia-to-curb-conspiracy-theory-videos>
- [28] Tom Simonite. 2019. Inside the Alexa-Friendly World of Wikidata. *WIRED* (Feb. 2019). <https://www.wired.com/story/inside-the-alexa-friendly-world-of-wikidata/>
- [29] Chris Tenove. 2020. Protecting democracy from disinformation: Normative threats and policy responses. *The International Journal of Press/Politics* 25, 3 (2020), 517–537. ISBN: 1940-1612 Publisher: SAGE Publications Sage CA: Los Angeles, CA.
- [30] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, and Faisal Azhar. 2023. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971* (2023).
- [31] Joshua Tucker, Andrew Guess, Pablo Barbera, Cristian Vaccari, Alexandra Siegel, Sergey Sanovich, Denis Stukal, and Brendan Nyhan. 2018. Social Media, Political Polarization, and Political Disinformation: A Review of the Scientific Literature. *SSRN Electronic Journal* (2018). doi:10.2139/ssrn.3144139
- [32] Nicholas Vincent, Isaac Johnson, Patrick Sheehan, and Brent Hecht. 2019. Measuring the Importance of User-Generated Content to Search Engines. *Proceedings of the International AAAI Conference on Web and Social Media* 13 (July 2019), 505–516. doi:10.1609/icwsm.v13i01.3248
- [33] Molly White. 2025. Elon Musk and the right's war on Wikipedia. *Citation Needed* (Jan. 2025). <https://www.citationneeded.news/elon-musk-and-the-rights-war-on-wikipedia/>
- [34] Wikimedia Foundation, Trust & Safety Disinformation team. 2021. The case of croatian wikipedia: encyclopaedia of knowledge or encyclopaedia for the nation? https://meta.wikimedia.org/wiki/File:Croatian_WP_Disinformation_Assessment_-_Final_Report_EN.pdf
- [35] Joe Wilkins. 2025. Leaked Documents Show Plot to Dox Anonymous Wikipedia Editors Using Hacking and Facial Recognition. *Futurism* (Jan. 2025).
- [36] Kate Wright, Martin Scott, and Melanie Bunce. 2024. *Capturing News, Capturing Democracy: Trump and the Voice of America*. Oxford University Press.
- [37] Kamy Yadav. 2021. *Platform Interventions: How Social Media Counters Influence Operations*. Technical Report. <https://carnegieendowment.org/posts/2021/01/platform-interventions-how-social-media-counters-influence-operations?lang=en>